DSE3140 • COMPUTER VISION
SUMMATIVE RESEARCH PROJECT

A Project Report

on

PNEUMONIA CLASSIFICATION FROM PEDIATRIC CHEST X-RAYS

A comparison of a baseline CNN, frozen MobileNetV2, and fine-tuned MobileNetV2

carried out as part of DSE3140: Computer Vision

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ABSTRACT

This report presents a three-model comparison for binary classification of pediatric chest X-ray images as Normal or Pneumonia. The experiment uses PneumoniaMNIST from the MedMNIST v2 benchmark at a configured image size of 224 × 224 pixels. The three configurations are a baseline convolutional neural network (CNN) trained from scratch, MobileNetV2 with its ImageNet-pretrained feature extractor frozen, and MobileNetV2 after fine-tuning selected pretrained feature layers. All models are evaluated on the same 624-image held-out test split, which contains 234 Normal and 390 Pneumonia-labelled images.

At the default threshold of 0.5, the fine-tuned MobileNetV2 achieved the strongest overall balance among the evaluated models: accuracy 0.885, precision 0.850, F1-score 0.915, specificity 0.709, and ROC-AUC 0.983. The frozen MobileNetV2 produced the highest recall (0.992), while the fine-tuned configuration reduced false-positive Normal cases from 92 to 68 relative to the frozen model. The report also examines bootstrap confidence intervals, paired differences, validation-selected thresholds, training histories, and an exploratory temperature-scaling analysis. These results describe a single educational benchmark experiment; they do not establish clinical validity or generalization to other hospitals or patient populations.

Keywords: pediatric chest X-ray, pneumonia classification, PneumoniaMNIST, convolutional neural network, transfer learning, MobileNetV2, fine-tuning, ROC-AUC, calibration.

REPORT ORGANIZATION

The report is organized to document the dataset, model design, experimental procedure, quantitative evaluation, responsible-use considerations, and opportunities for further study. The sections below provide a navigation overview; page positions may change when the document is opened in another Word-compatible application.

1. Introduction and project objectives
2. Background and related work
3. Dataset and preprocessing
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6. Implementation and reproducibility
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1. INTRODUCTION

Chest radiography is commonly used as part of the clinical assessment of respiratory illness, including suspected pneumonia in children. Interpreting a chest radiograph requires medical expertise and consideration of patient history, symptoms, image quality, and other evidence. Deep-learning models can learn visual patterns from labelled images and produce a reproducible classification output for research, but an automated label cannot replace professional interpretation or the broader clinical process.

Convolutional neural networks (CNNs) are widely used in image classification because their convolutional filters learn local visual representations and combine them into higher-level features. Kermany et al. demonstrated the potential of image-based deep learning for pediatric pneumonia classification using chest radiographs [1]. Subsequent work has emphasized that performance depends not only on model architecture but also on dataset construction, preprocessing, evaluation design, and the representativeness of the images used for testing.

1.1 Problem statement

The project investigates how classification performance changes when a CNN trained from random initialization is compared with a pretrained MobileNetV2 model. The transfer-learning model is evaluated in two stages: first with the feature extractor frozen, and then after fine-tuning selected pretrained layers. The labels are Normal (class 0) and Pneumonia (class 1). The central question is whether learned ImageNet features and subsequent adaptation produce a more useful balance of sensitivity, specificity, and overall discrimination on a held-out PneumoniaMNIST test split.

1.2 Objectives

- Inspect the PneumoniaMNIST benchmark, its supplied partitions, and its Normal/Pneumonia class distribution.
- Implement a baseline CNN trained from scratch and two MobileNetV2 transfer-learning configurations.
- Apply model-specific preprocessing and training-only augmentation without augmenting validation or test images.
- Compare the three models using accuracy, precision, sensitivity/recall, specificity, F1-score, ROC-AUC, and confusion-matrix counts.
- Examine model histories, confidence intervals, paired metric differences, and operating thresholds selected using validation data.
- Document the results, reproducibility requirements, limitations, and responsible-use boundaries.

1.3 Scope and contribution

The contribution is a controlled educational comparison rather than a new clinical diagnostic method. All three configurations use the same benchmark partitions and held-out test images. The analysis goes beyond reporting a single accuracy value by examining class-specific errors and uncertainty. The experiment does not include external-hospital testing, prospective patient evaluation, clinical reader comparison, or regulatory assessment, so results should be interpreted only within the benchmark setting.

2. BACKGROUND AND RELATED WORK

2.1 CNNs for chest X-ray classification

A CNN applies learned filters to image regions and uses nonlinear activations and pooling or strided operations to build progressively more abstract features. For a binary classifier, the final representation is mapped to a single sigmoid score. In this study, a score at or above the chosen decision threshold is labelled Pneumonia. This score reflects the model output under its training and preprocessing setup; it should not automatically be interpreted as a calibrated probability of disease.

The baseline CNN is valuable because it provides a reference model whose weights are learned from the target training split. It also illustrates a common challenge of small or imbalanced datasets: learning from scratch can be sensitive to initialization, optimization, class composition, and augmentation. Therefore, a baseline helps measure whether transfer learning provides an empirical improvement for this specific dataset.

2.2 Transfer learning and MobileNetV2

Transfer learning begins with representations learned on a larger source dataset and adapts them to a target task. MobileNetV2 uses inverted residual blocks and linear bottlenecks and was designed to balance recognition performance with computational efficiency [2, 6]. In this experiment, the model starts from ImageNet-pretrained weights, its original classification head is replaced, and a binary head is trained for the Normal/Pneumonia task.

The frozen configuration keeps the pretrained convolutional base fixed and learns only the new classification head. This limits the number of trainable parameters and is often a practical first transfer-learning stage. Fine-tuning allows selected pretrained features to adapt to the target images. It can improve task-specific representations, but it also introduces additional optimization choices and may increase the risk of overfitting if not carefully controlled. The official TensorFlow transfer-learning guidance describes the distinction between feature extraction and fine-tuning [4].

2.3 Benchmark datasets and evaluation discipline

MedMNIST v2 provides standardized lightweight biomedical image-classification datasets for reproducible research and education [3]. PneumoniaMNIST offers a compact pediatric chest X-ray task with supplied train, validation, and test partitions. Its small, processed benchmark format is convenient for coursework experiments, but it does not represent the diversity of full-resolution images, scanners, institutions, or patient groups encountered in clinical practice.

Medical-imaging reporting guidance such as CLAIM encourages transparent descriptions of data sources, partitioning, model development, evaluation metrics, uncertainty, and potential bias [10]. This report follows that general principle by separating validation-based choices from test reporting, exposing confusion-matrix counts, and describing the incomplete evidence around calibration and generalization.

3. DATASET AND PREPROCESSING

3.1 Dataset source and labels

The experiment uses PneumoniaMNIST from the MedMNIST v2 collection [3, 5]. The dataset is based on pediatric chest X-rays from Guangzhou, China, and the recorded experiment configures the images at 224 × 224 pixels. Each sample has one of two labels: Normal (0) or Pneumonia (1). The supplied split counts are retained without generating a new random split, which allows the model comparisons to use the same held-out test set.

Table 3.1. Dataset split and class distribution.

Split	Total images	Normal	Pneumonia
Training	4,708	1,214	3,494
Validation	524	135	389
Test	624	234	390

The training split is imbalanced: 3,494 of 4,708 images (approximately 74.2%) are labelled Pneumonia, while 1,214 (approximately 25.8%) are Normal. A model that predicts the majority class too frequently can achieve deceptively reasonable accuracy while missing class-specific errors. The training procedure therefore uses class weights calculated from the training labels (approximately 1.939 for Normal and 0.674 for Pneumonia). Class weights alter the contribution of samples to the training loss; they do not increase the amount of independent data.

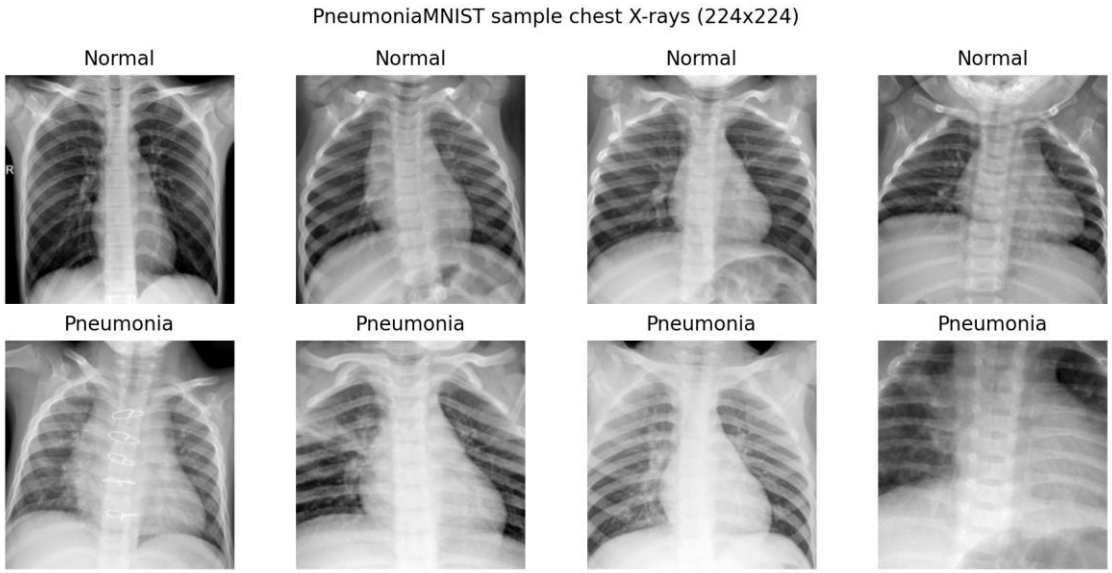


Figure 3.1. Representative 224 × 224 PneumoniaMNIST images: Normal samples are shown in the top row and Pneumonia samples in the bottom row.

3.2 Image preprocessing

The baseline CNN consumes a single-channel grayscale image. Pixel intensities are cast to floating point and scaled from the 0–255 range to approximately [0, 1]. MobileNetV2 expects a three-channel input; accordingly, the grayscale channel is repeated three times and transformed to the range expected by the pretrained network (approximately [−1, 1]). Consistent preprocessing is required at training and inference time, because a mismatch can change the distribution of the network input and degrade predictions.

3.3 Training-only augmentation

Training-time augmentation includes small random rotations, zoom, and contrast adjustments. The recorded configuration uses a rotation factor of 0.05, zoom factor of 0.1, and contrast factor of 0.1. Horizontal flips were not used in the original data-preparation design. Augmentation is applied only to training data so that validation and test metrics are computed from the supplied images rather than transformed variants. Keeping evaluation data free of random augmentation also makes comparisons between model configurations easier to interpret.

The source material does not establish patient-level independence or whether any duplicate or closely related image instances appear across the official partitions. These are relevant checks for future work, because image-level separation alone does not necessarily prove that test images are independent of training images at the patient level.

4. METHODOLOGY AND EXPERIMENTAL SETUP

4.1 Experimental workflow

1. Load the dataset through the MedMNIST interface and inspect split sizes, labels, and representative images.
2. Apply consistent image-size handling and the preprocessing pipeline required by each model.
3. Apply augmentation and class weighting to the training samples only.
4. Train the baseline CNN, the frozen MobileNetV2 classifier, and the fine-tuned MobileNetV2 configuration.
5. Monitor validation loss and preserve the selected checkpoint/history for each training run.
6. Generate test predictions for all three models on the same held-out test split.
7. Compute classification metrics, confusion matrices, ROC curves, bootstrap intervals, paired comparisons, threshold analysis, and the exploratory calibration summary.

4.2 Baseline CNN trained from scratch

The baseline receives a 224 × 224 × 1 input. Four convolutional blocks use 32, 64, 128, and 128 filters, respectively. Each block uses a 3 × 3 convolution with same padding and ReLU activation, followed by max pooling. Global average pooling reduces the spatial feature maps to a vector, after which a 128-unit dense layer, dropout with rate 0.5, and a one-unit sigmoid layer produce the classification score. The recorded model contains approximately 256,897 trainable parameters.

This architecture is compact enough for the coursework environment and provides a direct baseline for comparing transfer learning. Since the CNN starts from random initialization, its filters must learn task-specific visual representations from the relatively small training partition. The updated history plots show substantially more variation in validation loss and accuracy than the MobileNetV2 runs, indicating that this particular run had less stable validation behaviour.

4.3 Frozen MobileNetV2

The frozen MobileNetV2 begins with ImageNet-pretrained weights and excludes the original classification head. The convolutional base is kept frozen while a task-specific head is trained. The head uses global average pooling, dropout with rate 0.3, and a single sigmoid output. The recorded architecture contains 2,259,265 total parameters, of which 1,281 are trainable and 2,257,984 are frozen. The main benefit of this configuration is that it can reuse pretrained visual features while updating comparatively few weights on the target dataset.

4.4 Fine-tuned MobileNetV2

The fine-tuned configuration starts from the same pretrained MobileNetV2 family but allows selected pretrained feature layers to be updated during an additional training stage. Fine-tuning provides a way to adapt general image features to the texture and structure of pediatric chest X-rays. It can improve target-task discrimination, although the choice of layers, learning rate, and training duration can affect stability. The available exported results identify this configuration as fine-tuned but do not provide enough detail to state the exact number of unfrozen layers; this report therefore does not invent that value.

4.5 Training configuration

Table 4.1. Main training configuration recorded for the experiments.

Setting	Recorded value
Input image size	224 × 224 pixels; grayscale source
Batch size	32
Random seed	42
Optimizer / loss	Adam / binary cross-entropy
Recorded main-phase learning rate	1×10^{-3}
History length	CNN: 6 epochs; frozen MobileNetV2: 10 epochs; fine-tuned MobileNetV2: 15 epochs
Checkpoint selection	Validation-loss monitoring; early stopping/restoration as recorded in the notebook
Runtime environment	Google Colab GPU runtime (reported as T4), TensorFlow/Keras
Recorded wall-clock training time	CNN: 59.0 s; frozen MobileNetV2: 98.3 s; fine-tuned MobileNetV2: 164.8 s

The reported runtime values are the elapsed times recorded for these runs and depend on the runtime environment, compilation/caching effects, and implementation details. They should not be interpreted as hardware-independent speed benchmarks. The saved history lengths are reported as observed outputs rather than as a general guarantee of how many epochs an independently rerun experiment will execute.

4.6 Evaluation measures

Pneumonia is treated as the positive class. True positives (TP) are Pneumonia images predicted as Pneumonia; true negatives (TN) are Normal images predicted as Normal; false positives (FP) are Normal images predicted as Pneumonia; and false negatives (FN) are Pneumonia images predicted as Normal. Predictions at the default operating point use a threshold of 0.5. Accuracy measures overall correctness; precision measures the fraction of Pneumonia predictions that are correct; recall or sensitivity measures the fraction of labelled Pneumonia cases detected; specificity measures the fraction of Normal cases correctly identified; and F1-score balances precision and recall. ROC-AUC measures ranking discrimination across thresholds, rather than performance at a single threshold [7].

The core definitions are: Accuracy = (TP + TN)/(TP + TN + FP + FN); Precision = TP/(TP + FP); Recall = TP/(TP + FN); Specificity = TN/(TN + FP); and F1 = 2 × Precision × Recall/(Precision + Recall). Metrics serve different purposes, so the report uses them together rather than choosing a model from accuracy alone.

5. RESULTS AND ANALYSIS

5.1 Held-out test results at threshold 0.5

Table 5.1 summarizes performance on the same 624-image test partition. The fine-tuned MobileNetV2 achieved the strongest test accuracy (0.885), precision (0.850), F1-score (0.915), specificity (0.709), and ROC-AUC (0.983). The frozen MobileNetV2 achieved the highest recall (0.992), missing only three Pneumonia-labelled images. The fine-tuned model missed four such images, but it correctly identified more Normal samples, reducing false positives from 92 to 68 relative to the frozen configuration.

Table 5.1. Held-out test performance at the default classification threshold of 0.5.

Model	Accuracy	Precision	Recall	Specificity	F1-score	ROC-AUC	Time (min)
CNN (scratch)	0.803	0.785	0.944	0.568	0.857	0.893	0.98
MobileNetV2 frozen	0.848	0.808	0.992	0.607	0.891	0.970	1.64
MobileNetV2 fine-tuned	0.885	0.850	0.990	0.709	0.915	0.983	2.75

Table 5.2. Confusion-matrix counts on the held-out test set.

Model	True negatives (TN)	False positives (FP)	False negatives (FN)	True positives (TP)
CNN (scratch)	133	101	22	368
MobileNetV2 frozen	142	92	3	387
MobileNetV2 fine-tuned	166	68	4	386

Compared with the CNN baseline, the frozen model improved accuracy by 0.045 and ROC-AUC by 0.077. Fine-tuning then improved accuracy by a further 0.037 and ROC-AUC by 0.013 relative to the frozen model. The strongest measured model is therefore not identical on every metric: the frozen model leads in recall, whereas fine-tuning produces higher specificity and fewer false-positive Normal cases. That distinction matters because a screening-oriented operating point may prioritize missed Pneumonia cases differently from an application that also wishes to reduce false alarms.

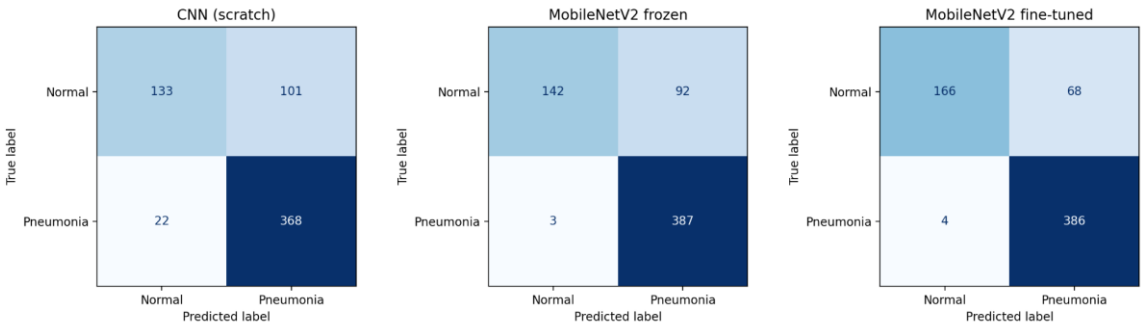


Figure 5.1. Confusion matrices on the held-out test set. Rows represent true labels and columns represent predicted labels.

The confusion matrices show that all models detect many positive cases, but they differ in how they treat Normal images. The baseline CNN misclassifies 101 of 234 Normal images as Pneumonia, frozen MobileNetV2 misclassifies 92, and the fine-tuned model misclassifies 68. Specificity rises from 0.568 to 0.607 and then to 0.709. This is a meaningful change in the balance between classes, but 68 false positives remain substantial and preclude any claim that the model is suitable for clinical use.

5.2 ROC analysis

The ROC curves in Figure 5.2 compare true-positive rate against false-positive rate across decision thresholds. Fine-tuned MobileNetV2 achieved the highest ROC-AUC (0.983), followed by frozen MobileNetV2 (0.970) and the baseline CNN (0.893). The higher area suggests that the fine-tuned scores rank positive cases above negative cases more effectively on this test set across thresholds. ROC-AUC does not, by itself, establish probability calibration, clinical utility, or transfer to a different source population.

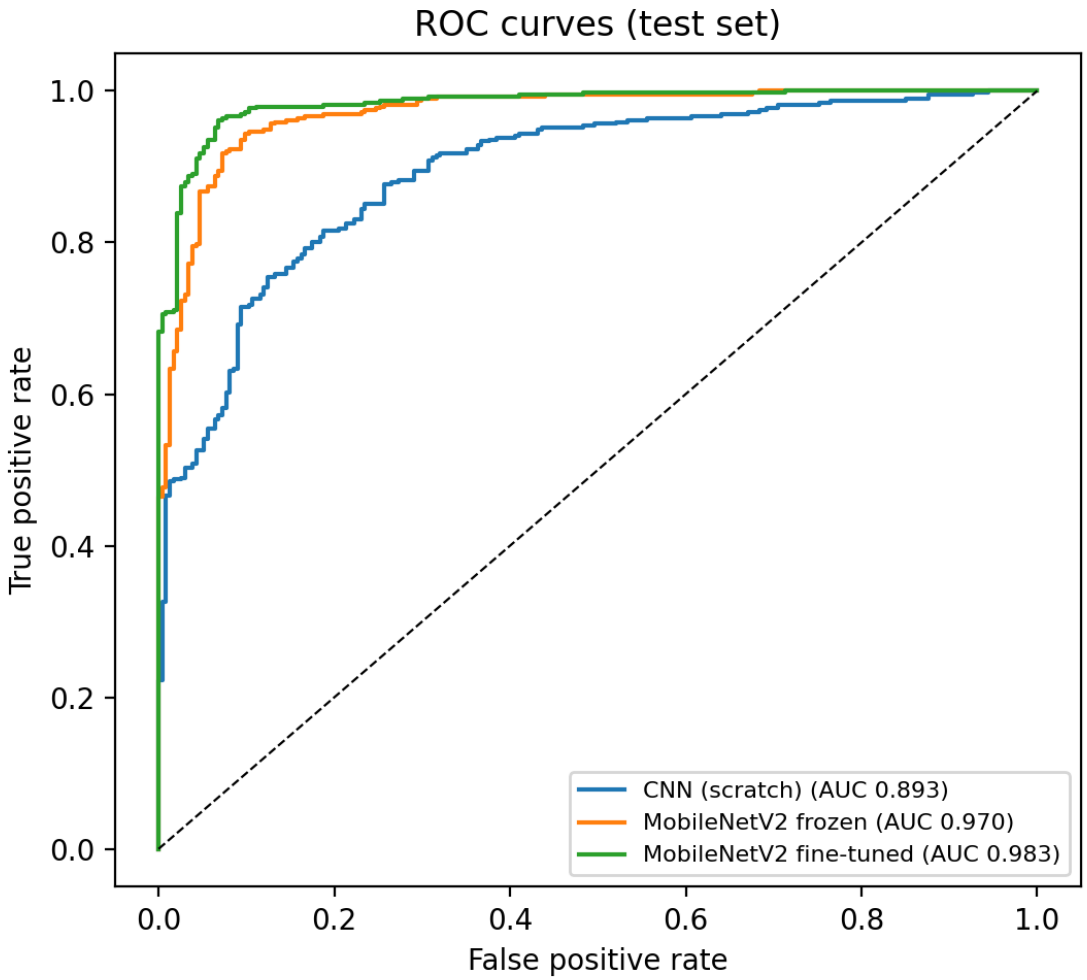


Figure 5.2. ROC curves for the three models on the 624-image held-out test set.

5.3 Training behaviour

Figures 5.3–5.5 show the saved training and validation histories for each model. The CNN training loss decreases overall, but validation loss is highly variable and rises sharply near the final recorded epoch; validation accuracy also fluctuates and falls to approximately 0.525 at the final recorded point. This behaviour indicates that the baseline run is unstable on validation data and should be examined with repeated runs and careful checkpoint verification.

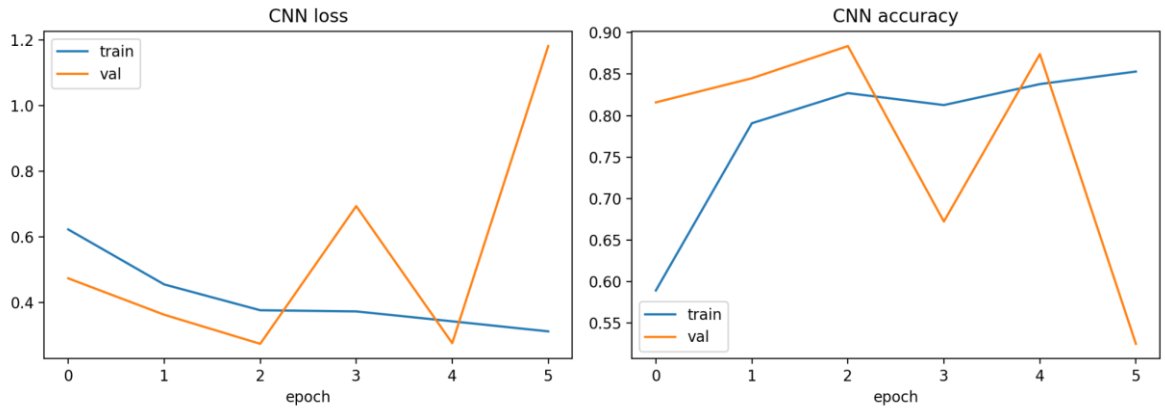


Figure 5.3. Baseline CNN training and validation loss and accuracy histories.

The frozen MobileNetV2 curves show a steady reduction in loss and high validation accuracy after the early epochs. The fine-tuned MobileNetV2 curves also show gradual reduction in training loss and validation loss that remains low across the recorded history. Its validation accuracy is generally close to 0.97–0.98. These curves support the conclusion that transfer learning produced more stable optimization behaviour in these particular runs, although one run per configuration cannot characterize variation across random seeds.

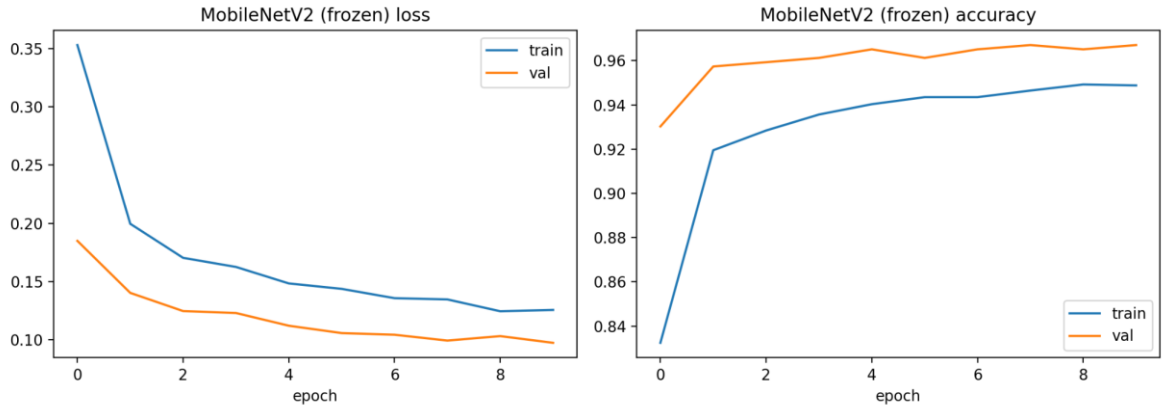


Figure 5.4. Frozen MobileNetV2 training and validation loss and accuracy histories.



Figure 5.5. Fine-tuned MobileNetV2 training and validation loss and accuracy histories.

5.4 Validation-set behaviour

Validation results are useful for model selection and for choices such as operating-threshold selection. They are not a substitute for held-out test performance. The recorded validation summary shows substantially higher accuracy and AUC for both MobileNetV2 configurations than for the baseline

CNN. Fine-tuned MobileNetV2 achieved validation accuracy 0.979 and ROC-AUC 0.997, with sensitivity 0.992 and specificity 0.941 at the default threshold. The corresponding baseline CNN validation accuracy and ROC-AUC were 0.884 and 0.941.

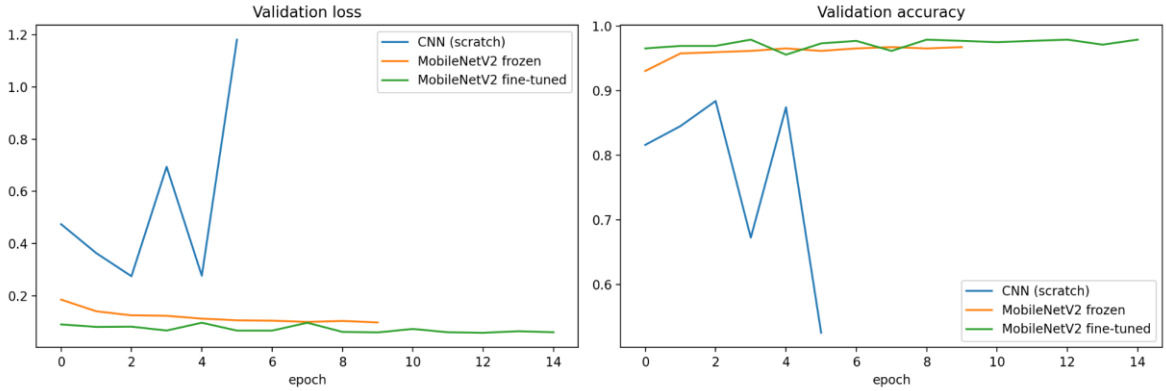


Figure 5.6. Validation loss and accuracy histories compared across all three model configurations.

Table 5.3. Validation-set summary at the default threshold of 0.5.

Model	Accuracy	ROC-AUC	Sensitivity	Specificity	FN	FP
CNN (scratch)	0.884	0.941	0.903	0.830	37	23
MobileNetV2 frozen	0.967	0.994	0.987	0.911	5	12
MobileNetV2 fine-tuned	0.979	0.997	0.992	0.941	3	8

The baseline CNN's validation behaviour is particularly uneven, and the validation results are stronger than the final test metrics for all configurations. The gap warrants further investigation rather than an assumption that a specific cause has been proven. Differences in split composition, source characteristics, or image-level dependencies could contribute; confirming these possibilities requires examination of the data construction and repeated evaluation.

5.5 Bootstrap confidence intervals

Bootstrap confidence intervals provide an estimate of uncertainty under a resampling procedure applied to the test results. They do not represent uncertainty across hospitals, datasets, training seeds, or clinical populations. Table 5.3 reports the supplied 95% intervals for accuracy, recall, F1-score, and ROC-AUC. The fine-tuned model has the highest point estimate for accuracy, F1-score, and AUC. The recall intervals for frozen and fine-tuned MobileNetV2 are both near the upper boundary, which is consistent with their high sensitivity in this test set.

Table 5.4. Test metrics with the available 95% bootstrap confidence intervals.

Model	Accuracy [95% CI]	Recall [95% CI]	F1 [95% CI]	ROC-AUC [95% CI]
CNN (scratch)	0.803 [0.772, 0.832]	0.944 [0.920, 0.966]	0.857 [0.833, 0.880]	0.893 [0.869, 0.916]
MobileNetV2 frozen	0.848 [0.819, 0.877]	0.992 [0.982, 1.000]	0.891 [0.869, 0.912]	0.970 [0.957, 0.981]
MobileNetV2 fine-tuned	0.885 [0.859, 0.910]	0.990 [0.978, 0.998]	0.915 [0.895, 0.934]	0.983 [0.974, 0.990]

5.6 Paired metric comparisons

The paired comparison evaluates model differences on matching test observations rather than treating the model results as if they came from unrelated test sets. The supplied interval for frozen MobileNetV2 relative to the baseline CNN is positive for accuracy, recall, F1-score, and ROC-AUC. When fine-tuned MobileNetV2 is compared with frozen MobileNetV2, the confidence intervals are positive for accuracy, F1-score, and ROC-AUC. The paired recall difference is −0.003 with an interval from −0.008 to 0.000; as supplied, the interval reaches zero and does not establish an unambiguous recall improvement for fine-tuning.

Table 5.5. Paired differences in test metrics (difference and 95% confidence interval).

Metric	Frozen MobileNetV2 − CNN	Fine-tuned − Frozen MobileNetV2
Accuracy	+0.045 [+0.016, +0.074]	+0.037 [+0.019, +0.056]
Recall	+0.049 [+0.028, +0.070]	−0.003 [−0.008, +0.000]
F1-score	+0.034 [+0.015, +0.054]	+0.024 [+0.012, +0.037]
ROC-AUC	+0.077 [+0.057, +0.098]	+0.013 [+0.007, +0.019]

5.7 Validation-selected threshold analysis

The default threshold of 0.5 provides a common baseline for comparing the three models, but it is not necessarily the only useful operating point. In a separate analysis, one threshold per model was selected on validation data to target validation specificity of approximately 0.963. These thresholds—0.777 for the CNN, 0.694 for frozen MobileNetV2, and 0.592 for fine-tuned MobileNetV2—were then applied to test predictions without selecting them from the test set. Raising a threshold generally reduces false-positive predictions but can also reduce sensitivity.

Table 5.6. Held-out test performance at the default and validation-selected thresholds.

Model	Val. threshold	Accuracy @0.5	Accuracy @selected	Sensitivity @selected	Specificity @0.5	Specificity @selected	FN @selected	FP @selected
CNN (scratch)	0.777	0.803	0.819	0.864	0.568	0.744	53	60
MobileNetV2 frozen	0.694	0.848	0.883	0.982	0.607	0.718	7	66
MobileNetV2 fine-tuned	0.592	0.885	0.888	0.987	0.709	0.722	5	65

For fine-tuned MobileNetV2, the validation-selected threshold increases test accuracy modestly from 0.885 to 0.888 and specificity from 0.709 to 0.722. Sensitivity decreases slightly from 0.990 to 0.987, while false negatives increase from four to five and false positives decrease from 68 to 65. For frozen MobileNetV2, the selected threshold increases test accuracy from 0.848 to 0.883 and specificity from 0.607 to 0.718, with sensitivity changing from 0.992 to 0.982. The CNN also gains specificity, but with a much larger sensitivity reduction. A threshold choice should reflect the intended use and error costs; no operating point is clinically endorsed by this educational study.

5.8 Calibration analysis

Calibration asks whether predicted scores correspond to observed event frequencies. A model can rank examples well while producing overconfident or underconfident scores. Temperature scaling is a post-processing approach that rescales logits by a fitted temperature parameter; calibration evaluation may use log-loss, Brier score, expected calibration error (ECE), and reliability diagrams [8, 9]. In general, lower values of these error measures are preferred, but their estimates depend on how the bins, samples, and calibration split are defined.

Table 5.7. Calibration metrics before and after temperature scaling in the exported summary.

Model	T	Log-loss before	Log-loss after	Brier before	Brier after	ECE before	ECE after
CNN (scratch)	0.84	0.4635	0.4934	0.1472	0.1510	0.1151	0.1259
MobileNetV2 frozen	0.71	0.3834	0.4744	0.1123	0.1201	0.1413	0.1435
MobileNetV2 fine-tuned	1.12	0.3702	0.3421	0.0908	0.0887	0.1048	0.1054

For fine-tuned MobileNetV2, the supplied summary reports reductions in log-loss (0.3702 to 0.3421) and Brier score (0.0908 to 0.0887), while ECE increases slightly (0.1048 to 0.1054). For the CNN and frozen MobileNetV2, all three reported values worsen after scaling. Thus, the summary does not show a consistent improvement across models. Critically, the exported calibration summary does not specify which split was used to estimate each temperature. Until this is verified, the numbers and figure should be treated as exploratory; a defensible protocol fits temperature on validation data only and applies it unchanged to the test predictions.

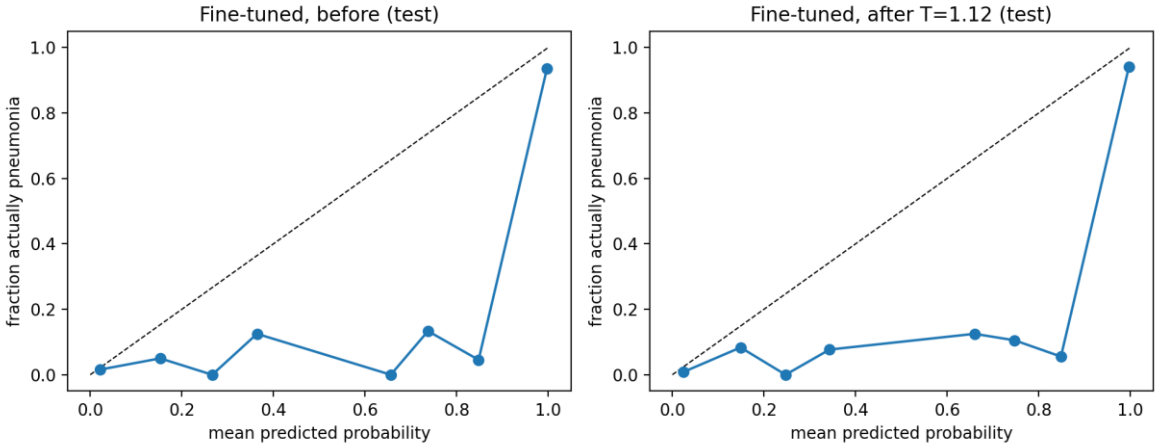


Figure 5.7. Fine-tuned MobileNetV2 reliability diagrams before and after temperature scaling (T = 1.12). Binned diagrams are diagnostic visualizations and do not alone establish clinical calibration.

6. IMPLEMENTATION AND REPRODUCIBILITY

The experiments are recorded in a Google Colab notebook implemented with Python and TensorFlow/Keras. The notebook contains the import and seed setup, dataset loading, input transformations, model definitions, training histories, test prediction generation, metric calculations, and visualization. The output files accompanying the report preserve the three-model test results, paired comparisons, bootstrap confidence intervals, threshold analysis, calibration summary, and figure exports. The most reliable reproducibility practice is to retain the executed notebook and the exact output files from the same run, rather than reconstructing results from copied fragments.

Table 6.1. Main software components and their roles.

Component	Purpose in the workflow
Python / Google Colab	Notebook execution and GPU-backed experiments
TensorFlow / Keras	CNN and MobileNetV2 model construction, training, saving, and inference
MedMNIST	Loading PneumoniaMNIST and its supplied data partitions
NumPy / pandas	Tensor operations, data handling, and tabular result management
scikit-learn	Classification metrics, ROC-AUC, confusion matrices, and evaluation utilities
Matplotlib / Seaborn	Training curves, ROC curves, confusion matrices, and calibration visualizations

6.1 Prediction procedure

1. Load the intended saved model and confirm that it matches the preprocessing pipeline used for that model.
2. Load an image and resize it to the configured 224 × 224 input dimensions.
3. Convert the image to the expected channel format and scale: [0, 1] for the CNN or approximately [−1, 1] three-channel input for MobileNetV2.
4. Run model inference to obtain the sigmoid score.
5. Apply the stated decision threshold (0.5 by default, or a separately documented validation-selected threshold).
6. Display the predicted label and score while clearly stating that the output is a coursework demonstration, not a medical diagnosis.

For reproducibility, record the exact model checkpoint, software versions, seed, preprocessing code, threshold, and evaluated split. When reporting a fine-tuned model, retain the details of which layers were unfrozen and the training-stage settings. The existing exported results identify the fine-tuned configuration but do not provide every one of these implementation details, so the final notebook should be checked before any attempt to replicate the run exactly.

7. LIMITATIONS AND RESPONSIBLE USE

7.1 Dataset representativeness and independence

The experiment relies on one prepared pediatric benchmark rather than images drawn independently from multiple hospitals. Its input resolution, image acquisition conditions, labels, and class composition may differ from those of a clinical workflow. The source material does not establish patient-level independence or the absence of duplicate/related images across the splits. Those checks should be completed before treating a test result as a reliable estimate of generalization.

7.2 Single-run and optimization uncertainty

Only one recorded training run is available for each configuration. Random initialization, data shuffling, GPU nondeterminism, and the exact early-stopping checkpoint can change the outcome. The bootstrap intervals summarize sampling uncertainty under the recorded resampling procedure, not variation across independently trained models or external data sources. Repeated runs with multiple seeds and more fully documented inference-time settings are necessary for a stronger model comparison.

7.3 Error trade-offs and threshold dependence

The fine-tuned model improves specificity relative to the frozen version but still flags 68 of 234 Normal test images as Pneumonia at threshold 0.5. The frozen model has the highest recall of the three configurations, while fine-tuning reduces false positives. This illustrates why one metric cannot fully characterize performance. Threshold selection changes the balance between false negatives and false positives; a threshold should be selected using validation data and its operating consequences evaluated on independent data before any deployment consideration.

7.4 Calibration uncertainty

The exported calibration results do not identify the split used to fit the temperature parameters. If the test set had been used for calibration fitting, the post-scaling test metrics would not be an unbiased final evaluation. The reported calibration section therefore remains provisional until the notebook confirms a validation-only fitting procedure. The calibration figure is a binned diagnostic, not proof that a score such as 0.9 represents a 90% chance of disease.

7.5 Clinical and ethical boundaries

No external validation, prospective evaluation, clinical reader study, demographic subgroup evaluation, prospective workflow trial, or regulatory assessment is included. The dataset is restricted to pediatric examples and does not establish performance for adults, other age groups, scanners, institutions, or patient populations. The project must not be used for diagnosis, triage, or treatment decisions. Any future application would require appropriate independent datasets, domain-expert review, clinical validation, risk assessment, privacy protections, and applicable ethical and regulatory review. Medical-imaging reporting guidance such as CLAIM highlights the importance of transparent methodology and uncertainty reporting [10].

8. CONCLUSION AND FUTURE WORK

This updated report compares a baseline CNN trained from scratch with frozen and fine-tuned MobileNetV2 configurations for binary classification of PneumoniaMNIST chest X-rays. On the held-out test set at threshold 0.5, fine-tuned MobileNetV2 achieved the best accuracy (0.885), precision (0.850), F1-score (0.915), specificity (0.709), and ROC-AUC (0.983). Frozen MobileNetV2 produced the highest recall (0.992), compared with 0.990 for the fine-tuned model. Fine-tuning reduced false positives from 92 to 68 compared with the frozen model, while the CNN baseline had the lowest specificity and most unstable recorded validation history.

The confidence intervals and paired comparisons reinforce the observed benefit of the transfer-learning configurations over the baseline for the reported metrics. They do not resolve limitations associated with the benchmark size, one run per model, unknown patient-level independence, or external generalization. The separate validation-selected threshold analysis showed that the fine-tuned model's test accuracy and specificity increased modestly at threshold 0.592, with a small reduction in sensitivity. Calibration results are not definitive until the temperature-fitting split is confirmed.

8.1 Recommended future work

- Repeat all three training configurations with multiple independent random seeds and report mean, standard deviation, and interval estimates.
- Verify the data-partition protocol, patient-level independence, and possible duplicate or closely related images across the supplied splits.
- Document the fine-tuning procedure precisely, including trainable layers, learning rates per stage, checkpoint selection, and software versions.
- Choose operating thresholds using validation data only and evaluate them on untouched test data; state the clinical or task-specific rationale for the sensitivity/specificity trade-off.
- Fit calibration parameters on a validation/calibration partition only, then evaluate calibration on untouched test data using multiple metrics and reliability plots.
- Evaluate on a genuinely external dataset and assess subgroup performance before making generalization claims.
- Explore explainability approaches such as Grad-CAM as research aids, with domain experts judging whether the highlighted regions are meaningful [11].
- Follow transparent medical-imaging AI reporting practices, including clear description of data, reference labels, model selection, uncertainty, and limitations [10].

APPENDIX A. RECOMMENDED CODE ORGANIZATION

For a reproducible project archive, organize the executed notebook into these logical sections: (1) imports and environment/version capture; (2) seed and determinism setup; (3) dataset loading and split verification; (4) sample inspection; (5) preprocessing and class weights; (6) augmentation; (7) CNN definition and training; (8) frozen MobileNetV2 definition and training; (9) fine-tuning stage and checkpoint selection; (10) held-out test predictions; (11) metric, bootstrap, and paired-difference calculations; (12) validation-selected threshold analysis; (13) calibration fitted on validation data only and evaluated unchanged on test data; and (14) plots and machine-readable result exports.

Before submission, verify that the notebook, model checkpoints, CSV summaries, confusion matrices, ROC curves, and histories all correspond to the same final run. In particular, update the calibration documentation with the precise split used to fit temperature scaling, and record the exact fine-tuning layers and parameters. This makes the report easier to audit and prevents figures and tables from silently referring to different model versions.

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